

EXECUTIVE REFERENCE PACKAGE · DECISION-READY

Stellar Intelligence

ASTRID — Astronomical System for Translating Radiance Into Discovery

Translating the life behind the light.

MVP Development · Pilot Adoption · Strategic Partnership

Prepared for executive review, STEM program stakeholders,
institutional partners, and product implementation teams.

NAVIGATE

Stellar Intelligence is an AI-powered astronomy research and observation platform. It combines object lookup, sky-map interaction, H-R diagram placement, image analysis, evidence trails, evolution timelines, telescope coaching, challenges, research briefs, and a persistent Stellar Journal. ASTRID is the universal reasoning agent that connects these modules into a single guided intelligence layer.

This package defines the product, the MVP, the agent architecture, the implementation logic, the commercial return model, and the next operational steps in a formal, investor- and institution-ready format.

PACKAGE NAVIGATION

\$	AREA	EXECUTIVE PURPOSE
01	Product Thesis	Defines the platform value and strategic premise.

02	ASTRID Intelligence Layer	Defines the agent reasoning standard and orchestration model.
03	MVP Module Blueprint	Specifies the feature set required for a credible MVP.
04	Data and Knowledge Base	Lists structured tables, source records, and app memory.
05	Generator and Button Logic	Connects visible UI actions to real outputs.
06	Scientific Integrity	Defines source labeling, uncertainty, and validation.
07	ROI and Commercial Model	Frames financial, educational, operational, and brand return.
08	90-Day MVP Roadmap	Outlines staged build and validation milestones.
09	Executive Call to Action	States the pilot ask and next activation steps.
10	Validation and Risk Controls	Protects credibility while supporting ambitious reasoning.
11	Implementation Appendix	Technical requirements for developers and builders.

Core Positioning

EXECUTIVE DASHBOARD — DECISION-READY CATEGORIES

ASTRID operates as a high-discernment scientific companion. She interprets user intent, reads app context, routes actions, retrieves or labels data, generates structured explanations, and guides the user toward the next useful step. She should feel conversational and human-centered, while remaining scientifically disciplined.

Reasoning Standard

01	Answer first, then organize supporting context.
02	Separate evidence, inference, discernment, reflection, and next action.
03	Do not expose hidden chain-of-thought; provide a concise user-facing reasoning summary.
04	Correct weak or unsupported claims directly but respectfully.
05	Use metaphysical or reflective framing only as interpretation, never as measurement or proof.
06	Offer action chips that connect to real modules: save, compare, evidence, timeline, report, sky map, image analysis, or telescope coach.

Intent and Action Router

NATURAL LANGUAGE ROUTED TO FUNCTIONAL APP BEHAVIOR

MODULE	CORE FUNCTION	SUCCESS OUTPUT
Ask ASTRID	Natural conversation, command routing, research guidance	Coherent answer plus suggested app actions
Research Agent	Object lookup for stars, systems, and exoplanets	Structured data profile with sources and confidence
Sky Map	Interactive object exploration	Object context, nearby targets, and ASTRID explanation
H-R Diagram	Clickable stellar placement and lifecycle context	Selected star profile and uncertainty-aware placement
Image Analysis	Upload and interpret sky images	Quality rating, object clues, possible matches
Evidence Trail	Source and reasoning transparency	Evidence cards with missing-data warnings
Evolution Timeline	Likely lifecycle pathway	Stage-based timeline and interpretation
Stellar Journal	Save research, observations, reflections	Persistent entry with next research question
Report Generator	Create exportable research briefs	Formatted Stellar Brief for user portfolio or review
STEM	Guided learning and	Feedback, completion status,

Challenges	assessment	and explanation
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Stellar Intelligence separates UI presentation from data, generated outputs, source records, saved journal entries, and agent context. This is the only way to make the product accurate, extensible, and testable.

Core Tables and Collections

RECOMMENDED DATA MODEL FOR MVP IMPLEMENTATION

TABLE / COLLECTION	PURPOSE	KEY FIELDS
AstronomyObjects	Stars, exoplanets, systems, and catalog objects	ObjectID · name · type · aliases · coordinates · source IDs
Stars	Stellar properties for H-R placement	temperature · luminosity · radius · mass · spectral class · distance
ResearchQueries	ASTRID research searches	query · timestamp · intent · user context · status

ResearchResults	Structured object lookup outputs	<code>object · summary · values · confidence · source notes</code>
SourceRecords	Source provenance tracking	<code>source · retrieval mode · URL/API · timestamp · data label</code>
ImageAnalysisResults	Uploaded image interpretation	<code>image · quality · detected objects · confidence · warnings</code>
EvidenceRecords	Source-based reasoning cards	<code>claim · evidence · confidence · missing data · uncertainty</code>
JournalEntries	Stellar Journal records	<code>date · object · notes · AI takeaway · evidence · next question</code>
StellarReports	Generated briefs	<code>summary · H-R placement · evidence trail · conclusion</code>
AgentRunLogs	ASTRID run tracking	<code>intent · routed action · status · output summary</code>

A premium-looking interface is not enough. The product becomes credible only when each action creates state change, generated content, saved records, or module navigation. ASTRID should be able to call these same functions conversationally.

Generator and Button Map

PREVENTS DECORATIVE-ONLY UI BY LINKING ACTIONS TO FUNCTIONS AND OUTPUTS

VISIBLE ACTION	GENERATOR / FUNCTION	OUTPUT
Run ASTRID Action	<code>routeAstridIntent(query, context)</code>	Routed module action and response
Research Object	<code>generateResearchResult(query)</code>	Structured object profile
Analyze Image	<code>analyzeUploadedImage(image, metadata)</code>	Image assessment and object clues
Open Evidence Trail	<code>generateEvidenceTrail(selectedObject)</code>	Evidence cards and uncertainty notes
Open Evolution Timeline	<code>generateEvolutionTimeline(selectedStar)</code>	Lifecycle timeline
Generate Stellar Report	<code>generateStellarReport(selectedObject, session)</code>	Report preview and saved record
Save to Stellar Journal	<code>generateJournalEntry(context)</code>	Persistent journal entry

Compare to Sun	<code>generateComparison(selectedObject, sun)</code>	Side-by-side comparison
Start Telescope Coach	<code>generateTelescopeCoach(inputs)</code>	Observing guidance
Submit Challenge	<code>generateChallengeFeedback(answer, context)</code>	Corrective feedback and completion status

Stellar Intelligence should not pretend certainty. It should show users how scientific knowledge is built: source by source, measurement by measurement, and confidence level by confidence level. This makes the product stronger and more trustworthy.

Trusted Astronomy Sources

SOURCE	BEST USE	IMPLEMENTATION NOTE
NASA Exoplanet	Exoplanets, host stars,	Use TAP/API where available;

Archive	Kepler/TESS-related records	label retrieved fields.
SIMBAD / CDS	Object identifiers, bibliography, measurements, aliases	Use as an object identification and source-history layer.
ESA Gaia DR3	Astrometry, parallax, proper motion, photometry	Use for star-position and distance-related context.
NASA SkyView	Multi-wavelength sky image context	Use for sky-image and map context where appropriate.
Internal Sample Dataset	MVP fallback and demos	Label clearly as sample, cached, inferred, or simulated.

Data Labeling Standard



The ROI of Stellar Intelligence is multidimensional. It creates educational value,

operational efficiency, portfolio value, institutional relevance, and commercial pathways. The platform positions its creator as a serious AI systems architect working at the intersection of scientific visualization, explainable AI, and human-machine communication.

ROI Framework

EDUCATIONAL

Improved Science Engagement

STEM literacy, astronomy comprehension, and evidence-based reasoning across student populations.

Challenge completion · Journal usage · Explanation comprehension

OPERATIONAL

Educator Time Savings

Reduces time spent creating explanations, reports, and reusable lesson assets for outreach teams.

Reports generated · Prep time reduced · Lesson assets created

INSTITUTIONAL

Museum, School & University Adoption

Supports STEM programs, museum interpretation, and university outreach programs.

Pilot conversions · Workshop adoption · Partnerships

COMMERCIAL

Multiple Revenue Streams

Freemium, school licensing, museum deployments, white-label, and workshop pathways.

[Paid users](#) · [License revenue](#) · [Deployment contracts](#)

CAREER / BRAND

AI Architecture Portfolio

Demonstrates advanced AI product architecture and scientific communication at professional level.

[Portfolio impact](#) · [Invitations](#) · [Applications](#) · [Partnerships](#)

STRATEGIC

Grant and Public Science Funding

Strong fit for NASA, NSF, and STEM-style education initiatives targeting science literacy.

[Grant applications](#) · [Public engagement metrics](#)

Revenue and Funding Paths

PATH	BEST FIT	NOTES
Premium Individual Tier	Enthusiasts, students, independent learners	Advanced reports, journal exports, deeper comparisons
School / Classroom License	K-12 STEM, universities, astronomy classes	Teacher dashboards and challenge packs
Museum / Planetarium Deployment	Public science institutions	Kiosk or guided visitor experience
STEM Workshop Package	Outreach events, fellowships, camps	Prebuilt activities, reports, and certificates

Grant-Funded Public Science Tool	NASA/STEM-style education initiatives	Strong fit for public engagement and science literacy
White-Label Astronomy Platform	Institutions and learning platforms	Custom branding and curriculum integration

PRIMARY ASK

Approve a 90-day MVP pilot for Stellar Intelligence, focused on validating the core reasoning loop: search or observe, analyze with ASTRID, map to the H-R diagram, show evidence, save to the Stellar Journal, and generate a research brief.

- Finalize MVP scope and no-dead-button standard.
- Implement ASTRID intent routing and module-aware action chips.
- Connect a sample star/exoplanet dataset, then add live API pathways.
- Build Stellar Journal persistence and Report Generator export.
- Run a guided demo using Vega, Sirius B, Kepler-22b, TRAPPIST-1, and an uploaded sky image.
- Prepare a pilot deck and stakeholder walkthrough.

Proceed with a 90-day MVP pilot to validate Stellar Intelligence as a premium AI astronomy reasoning platform for STEM education, public science engagement, and human-machine scientific exploration.

Stakeholder Activation

AUDIENCE	WHAT THEY CAN DO	WHAT THEY RECEIVE	STRATEGIC VALUE
STEM Programs	Pilot with students or workshops	Guided astronomy research workflows	Improved science engagement and portfolio outputs
Museums / Planetariums	Test visitor-facing ASTRID experiences	Interactive explanations and saved visitor journeys	Higher interpretive value and modern AI engagement
Universities	Use in outreach or introductory astronomy	Research briefs, evidence trails, and journal entries	Bridge between education, AI, and scientific literacy
Funders / Sponsors	Support pilot	Clear impact metrics and	Visible STEM and AI-literacy

	development and validation	public-facing innovation	return
Technical Partners	Connect APIs, sky maps, or data services	Integration showcase and product credit	Practical demonstration of explainable AI in science

Risk and Control Matrix

RISK	IMPACT	CONTROL
Dead-button UI	Damages trust and makes demo feel superficial	Every visible action must map to function, state, output, or navigation
False certainty	Misleads users and weakens scientific credibility	Use confidence labels, missing-data warnings, and source notes
Over-poetic	Turns science into vague metaphor	Separate evidence from reflection at all times

responses		
Weak data provenance	Undermines research use and credibility	Maintain SourceRecords and data labels throughout
Scope creep	Prevents MVP completion and delays pilot	Build the reasoning loop first, add advanced modules second
API instability	Live data may fail during demo	Use labeled cached/sample fallback data at all times

Success Metrics

Search-to-Journal Time

Time from object query to saved journal record.

Shorter time → workflow clarity

Report Export Rate

Share of sessions that generate a Stellar Brief.

Higher rate → perceived output value

Evidence Trail Usage

How often users open source reasoning cards.

Higher engagement → trust and learning depth

Challenge Completion

Completed STEM challenges per user.

Completion rate → education engagement

Repeat Journal Usage

Users returning to saved research entries.

Return rate → memory system value

Demo-to-Pilot Conversion

Stakeholders requesting a pilot after demo.

Conversion rate → institutional traction

Required State Variables

selectedObject, selectedStar, currentSkyMapTarget, uploadedImage, imageAnalysisResult, lastResearchResult, currentEvidenceTrail, currentTimeline, currentReport, currentJournalEntry, explanationMode, currentModule, astridIntent, astridContext, astridConfidence, suggestedActions, agentRunLog

Source Notes

SOURCE	NOTE
NASA Exoplanet Archive	Supports programmatic access through API/TAP services. Provides planetary systems, Kepler, TESS, and related data products.
SIMBAD / CDS	Used for astronomical object identification, measurements, bibliography, and aliases.
ESA Gaia DR3	Major stellar astrometry and photometry source for positions, parallaxes, proper motions, and related star data.
NASA SkyView	Provides virtual-observatory image context across multiple surveys and wavelengths.

Stellar Intelligence should feel like a premium astronomy intelligence system: scientifically rigorous, emotionally resonant, visually elegant, and operationally functional. ASTRID is the system that turns curiosity into research, evidence into understanding, and observation into memory.

